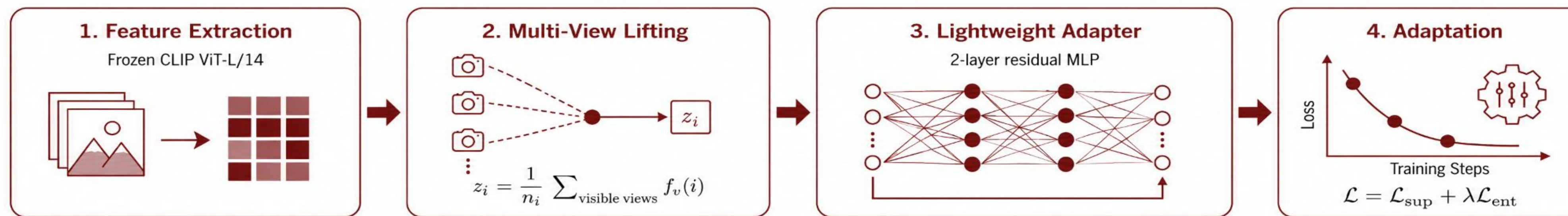


Optimization-Guided Adaptation for 3D Scene Understanding

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Pretrained 2D features are lifted into 3D once and frozen; only a lightweight adapter trained on sparse labels is optimized.

Overview

- Fused 2D visual features enable efficient 3D segmentation, but degrade in low-confidence or poorly covered regions.
- We test whether a lightweight adapter can improve segmentation under sparse labels without retraining a full 3D backbone.
- Our method freezes the lifted OpenScene/CLIP features and trains only a small residual MLP adapter.
- Goal:** label-efficient 3D adaptation with no end-to-end 3D retraining.

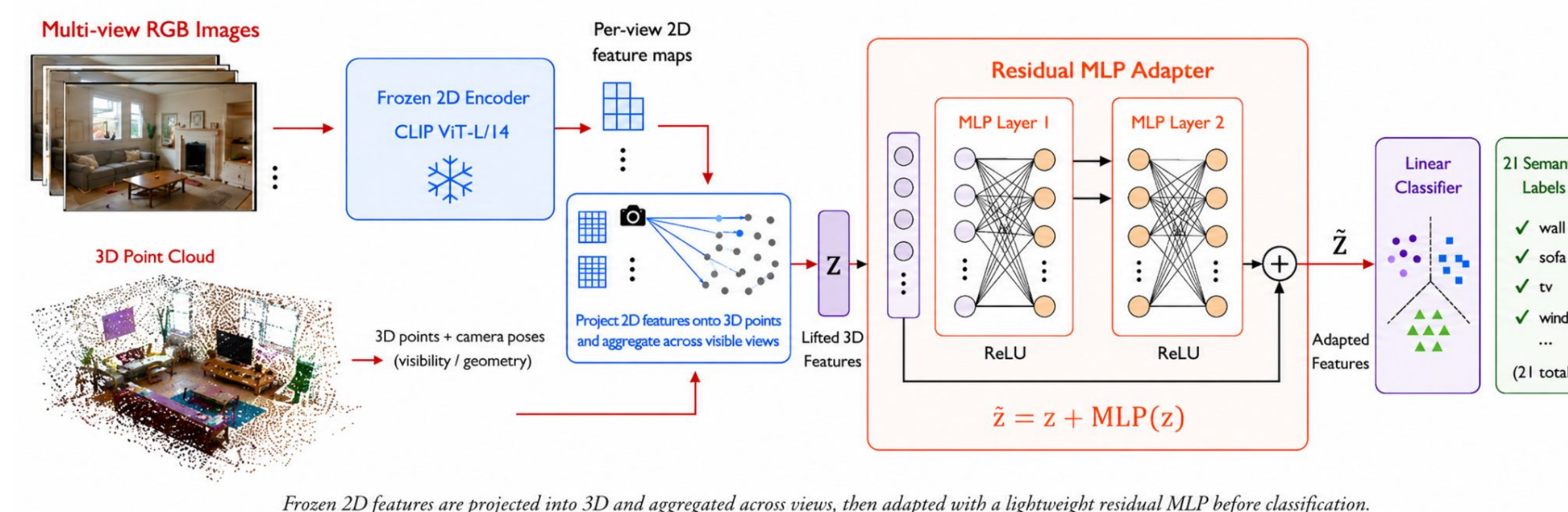
Dataset and Evaluation

- Dataset:** Matterport3D, 21 semantic classes
- Input:** voxelized 3D scenes with frozen OpenScene / CLIP lifted features
- Label setup:** 5% class-balanced labels per scene
- Evaluation:** scene-wise inference, aggregated over the test split
- Metrics:** mIoU

Hypotheses

- H1 — Boundary incoherence:** neighbors on the same surface receive inconsistent predictions.
- H2 — Under-coverage unreliability:** points seen in few views have higher-variance fused features.
- H3 — Adaptation drift:** training on tiny labeled sets can distort predictions in unlabeled regions.

Method



1) Feature Extraction:

Frozen CLIP ViT-L/14 -> 768-dim features.

2) Multi-View Lifting:

Project each 3D point into visible image views; average features:

$$z_i = \frac{1}{n_i} \sum_{v \in V(i)} f_v(i)$$

3) Lightweight Adapter:

2-layer residual MLP then linear classifier -> 21 classes

$$z_i \rightarrow \text{fc}_1 \rightarrow \text{LayerNorm} \rightarrow \text{ReLU} \rightarrow \text{fc}_2 \rightarrow \hat{z}_i$$

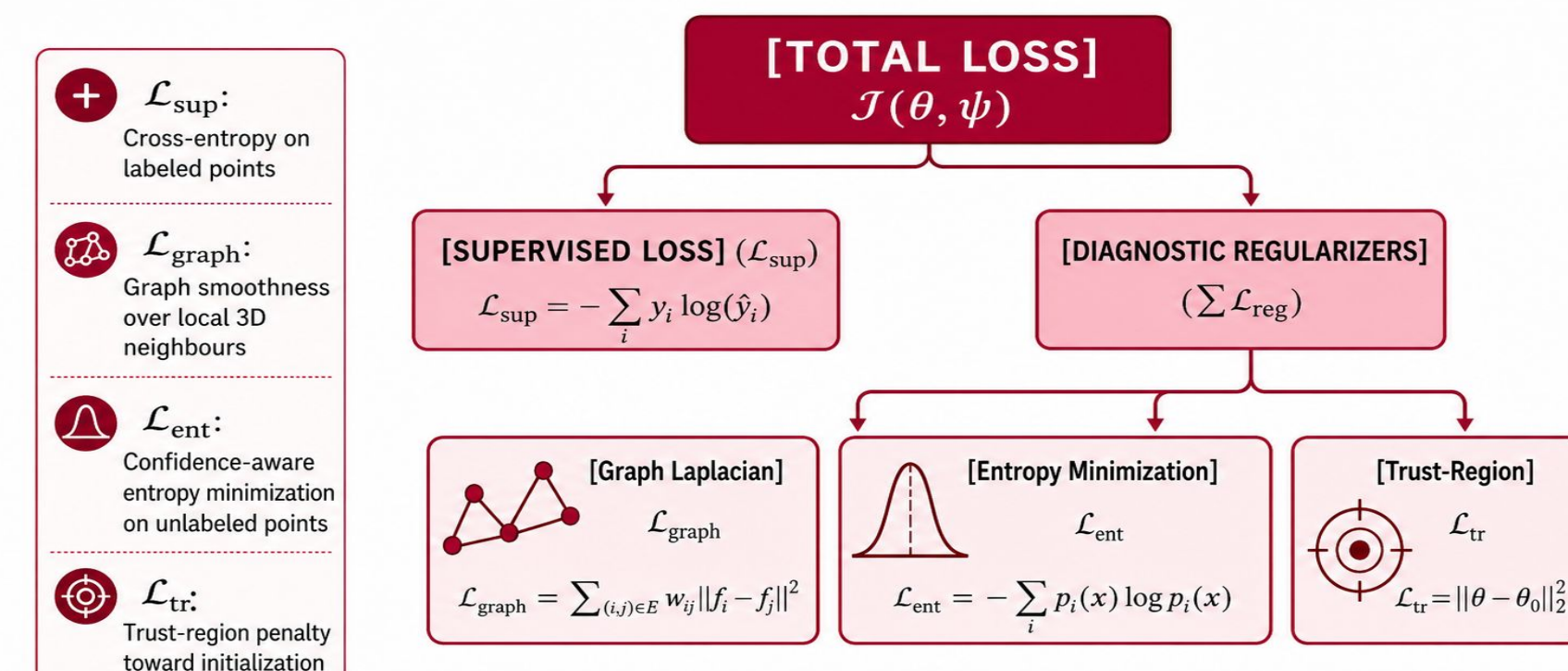
4) Optimization-Guided Adaptation:

train on ~5% class-balanced labels per scene; evaluate ablations:

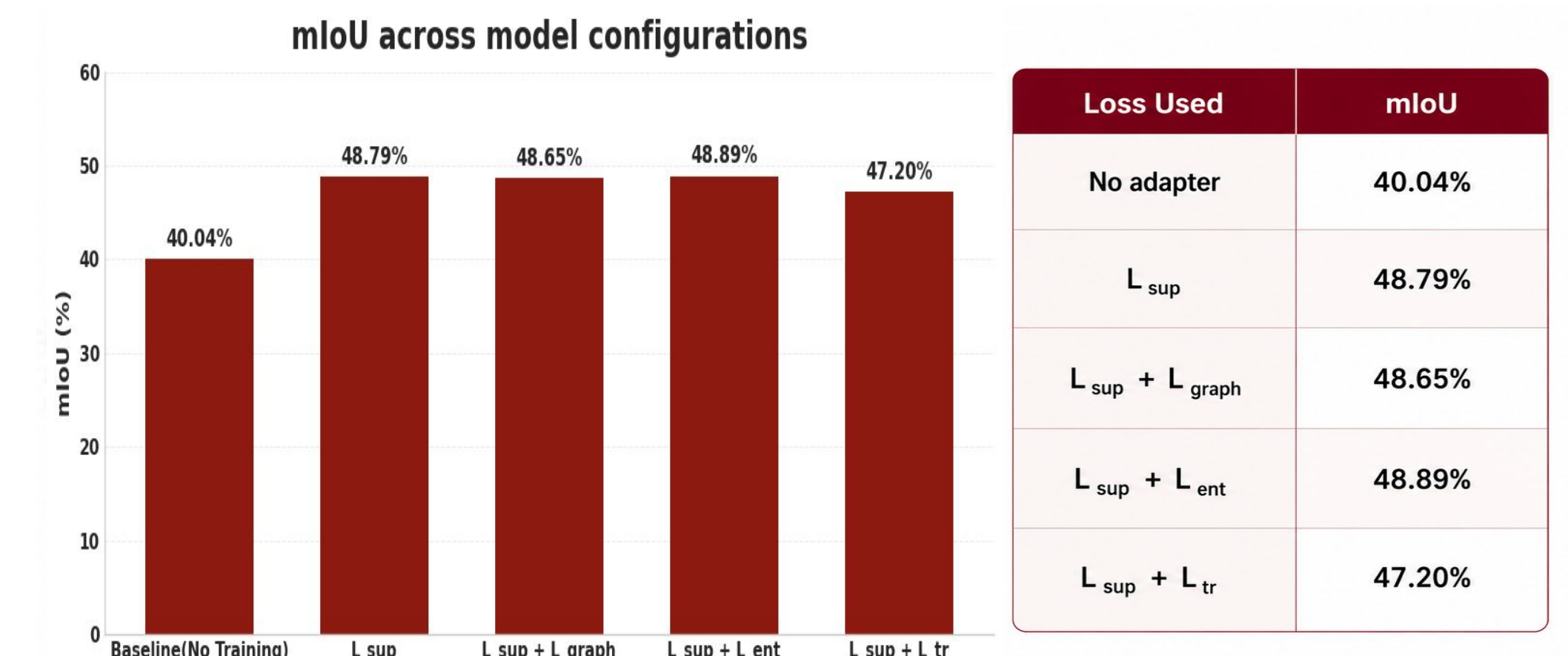
$$\mathcal{L}_{\text{sup}}, \mathcal{L}_{\text{sup}} + \lambda_1 \mathcal{L}_{\text{graph}}, \mathcal{L}_{\text{sup}} + \lambda_2 \mathcal{L}_{\text{ent}}, \mathcal{L}_{\text{sup}} + \lambda_3 \mathcal{L}_{\text{tr}}$$

Adapter / Objective

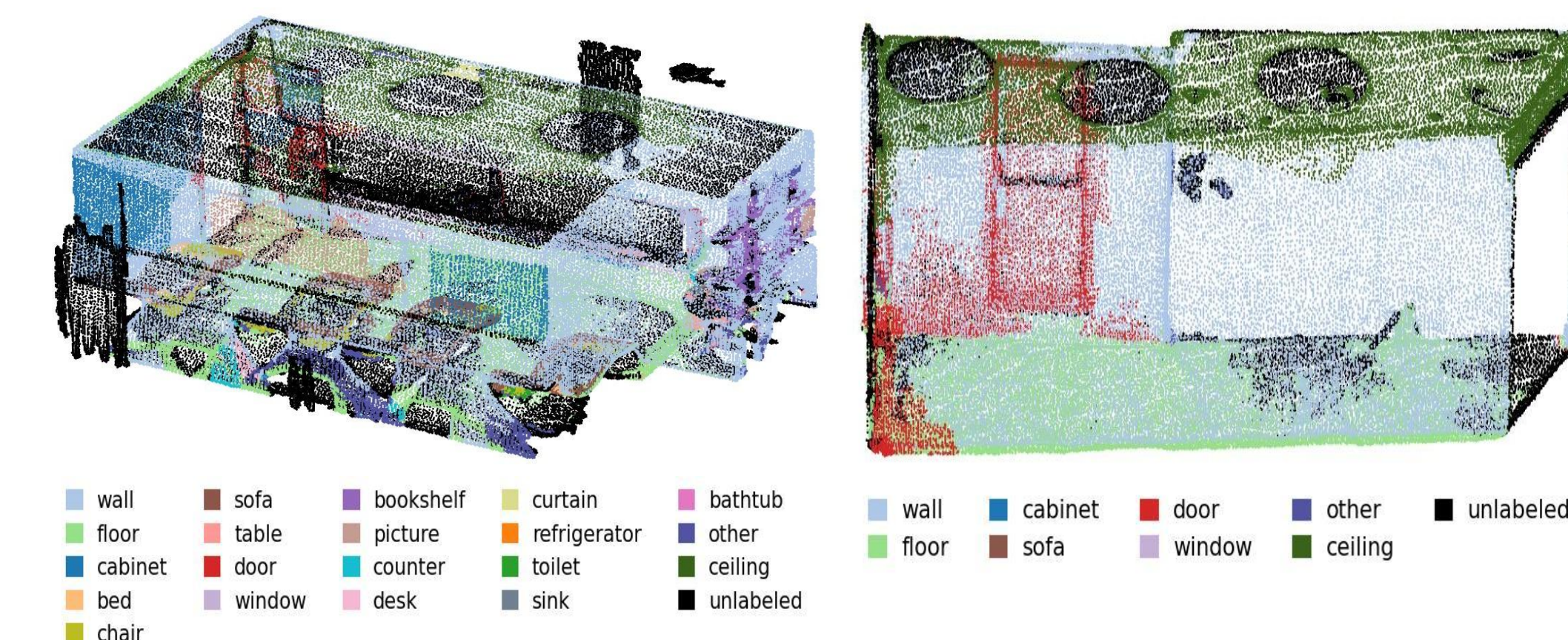
$$\text{TOTAL LOSS } \mathcal{J}(\theta, \psi) = \text{Supervised Loss } (\mathcal{L}_{\text{sup}}) + \sum \text{Regularization Losses } (w_i \cdot \mathcal{L}_{\text{reg}_i})$$



Results and Discussion



Segmented Scenes



- Strong overall gain:** mIoU improves from **40.04%** frozen baseline to **48.79%** with supervised adaptation.
- Entropy is marginally best:** $\mathcal{L}_{\text{sup}} + \mathcal{L}_{\text{ent}}$ reaches **48.89%**, only slightly above supervised-only adaptation.
- Regularizers have mixed effects:** graph smoothing is comparable, while trust-region underperforms in this setup.
- H2 remains a bottleneck:** low-view-coverage points stay difficult despite adaptation.

Conclusion and Future Work

- Lightweight adaptation:** improves frozen OpenScene / CLIP lifted features under 5% supervision.
- Best aggregate result:** entropy regularization gives the highest mIoU, but only marginally improves over supervised-only adaptation.
- Mixed regularizer effects:** graph smoothing is comparable, while trust-region underperforms in this setup.
- View-coverage bottleneck:** low-coverage points remain difficult, suggesting a structural limit from visibility rather than only the adaptation objective.
- Future work:** coverage-aware objectives, cross-dataset evaluation on ScanNet / Replica, and 3D UNet-based adapters.

Input Image Ground Truth

